DESIGN AND INNOVATION

ACADEMIC

Grade 11/12

Curriculum Guide

DESIGN AND INNOVATION

Prerequisite: None

Course Description:

This course develops learners' capacities to create meaningful community-driven responses through a human-centered, collaborative, and iterative design process to understand problems, create possibilities, build prototypes, and improve solutions with-users and communities. Using a Filipino-localized Design Thinking, learners strengthen *pakikipagkapwa,* empathy, listening, observation, problem finding, teamwork, and creative ideation. They also develop skills in prototyping, collecting feedback, making improvements, ethical thinking, and reflective communication.

Learners work in teams to complete a capstone project that meets a need in their school, community, workplace, or chosen field. The focus is not only on producing an innovative result, but on demonstrating a thoughtful design process: understanding people, reframing problems, testing assumptions, learning from feedback, and improving ideas with humility and care.

Elective: Academic

Time Allotment: 80 hours for one term

Design and Innovation

CONTENT DOMAIN	**CONTENT STANDARDS** *The learners demonstrate understanding of …*	**LEARNING COMPETENCIES** *The learners …*	**SUGGESTED OUTPUT**
Foundations of Human-Centered Design and Innovation *(Mga Batayan ng Disenyong Nakasentro sa Tao at Inobasyon)*	design and innovation as human-centered, collaborative, ethical, and iterative processes that respond to real needs and improve experiences.	1. explain design thinking as a human-centered, structured, non-linear, and iterative approach to solving real-world problems. 2. differentiate design thinking from other research-informed outputs such as project-making, research reporting, invention, and business planning. 3. describe the phases of the design thinking process: empathy, sense-making, ideation, prototyping, testing, and reflection. 4. analyze examples of real services, products, or other innovations that used the design thinking process. 5. identify the ethical responsibilities of designers when working with people and communities, including: a. obtaining informed consent; b. demonstrating respect for individuals and cultural diversity; c. safeguarding privacy and confidentiality; d. ensuring safety and inclusive participation; and e. preventing potential harm throughout the design process.	● Write a personal design reflection on the kind of designer they want to become and the values they uphold. ● Create an infographic explaining the design thinking process using local examples. ● Form design teams and establish norms for collaboration, feedback, documentation, and conflict resolution.

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Listening to Users and Understanding Contexts (*Himayin ang Konteksto)*	methods for understanding problems through the lived experiences, behaviors, needs, emotions, and environments of users and communities.	6. identify a possible area of concern in the school, community, workplace, or chosen field, using environmental scanning or other similar methods. 7. evaluate the team's current knowledge, underlying assumptions, gaps, and information needs related to the identified area of concern to guide further inquiry. 8. analyze the key users and stakeholders in the identified area of concern and the significance of their perspectives. 9. use ethical and respectful inquiry methods such as *kuwentuhan*, field observations, surveys, *anino*, and expert consultations to gather relevant information about key users and stakeholders, and the identified area of concern. 10. document users' stories, behaviors, challenges, constraints, and opportunities as bases for generating appropriate design solutions.	• Document prior knowledge using an inventory sheet or reflection journal. • Prepare a user/stakeholder map. • Combine and synthesize field notes, interview notes, photos/sketches, where appropriate. • Compile consent and ethics documentation. • Write field story summaries from the perspectives of users or community members.
Sensemaking and Problem Framing (*Himayin ang Konteksto)*	processes and mental models for organizing design research, identifying patterns, and reframing broad concerns into actionable design challenges.	11. analyze user data to identify patterns, tensions, needs, and opportunities derived from community stories and observations. 12. examine how different users experience a problem by considering their perspectives and reframing the situation from their points of view. 13. generate evidence-based insights explaining users' needs, challenges, and underlying reasons based on community data and context. 14. construct a design challenge or problem statement by synthesizing feedback from community	• Generate data synthesis organizers such as empathy maps, day-in-the-life narratives, *epiko*, and Rose-Thorn-Bud (RTB) technique. • Write insight statements. • Draft and revise a design challenge. • Create a problem-

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		members, peers, teachers, and experts. 15. explain why the selected design challenge or problem statement is meaningful, actionable, and well-defined based on the users' needs, insights, and context.	framing presentation.
Co-Design and ideation *(Ambagan ng mga Ideya)*	collaborative ideation methods that generate many possible responses to a design challenge before selecting ideas for testing.	16. generate multiple ideas to address the design challenge using different structured ideation methods, such as *bukas-daloy-sarado*, use cases and Substitute-Combine-Adapt-Modify-Out to another use-Eliminate-Reverse (SCAMPER) model. 17. facilitate collaborative sessions with users or community members identified area of concern. 18. develop solution ideas by building on, combining, and improving others' contributions during the collaborative ideation. 19. synthesize ideas to identify patterns, similarities, relationships, and promising directions for further development. 20. apply constructive feedback practices in critique sessions as both giver and receiver. 21. generate clear criteria such as context fit, feasibility, desirability, accessibility, ethics, safety, sustainability, and relevance to select, and refine ideas. 22. develop selected ideas into clear solution concepts responsive to user needs and contextual constraints.	• Document all ideas generated into an ideation wall or idea log. • Create a co-design or ideation session agenda and documentation. • Develop an idea selection matrix using team-generated criteria.
Making Ideas Testable *(Bumuo ng Prototype)*	prototyping as a quick, low-cost, visual, physical, or experiential way to test assumptions	23. explain the value of early prototyping in testing ideas and learning from feedback. 24. construct prototypes using available and appropriate materials. 25. identify assumptions to be tested through the	• Create low-fidelity prototypes such as sketches, storyboards, mockups, role plays, process maps, digital

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	and learn from users.	prototype. 26. gather preliminary feedback on prototypes from peers, teachers, and selected users. 27. document feedback and proposed improvements to refine the prototype and guide further development.	prototypes, or physical prototypes. ● Write an assumption-testing guide and feedback notes.
User Testing and Feedback gathering *(Ipakita, Suriin, at Ayusin)*	feedback gathering as a respectful, structured, and human-centered process for determining whether a prototype responds to actual needs and experiences.	28. conduct user or community testing through collaborative activities such as testing sessions, gallery walk, and *sama-sama scoring.* 29. observe how users interact with the prototype and document their actions, reactions, questions, points of confusion, and difficulties encountered. 30. collect feedback from diverse users and stakeholders regarding the usability, effectiveness, and relevance of the prototype, including those who experience the problem differently. 31. document feedback accurately without dismissing inconvenient or critical comments.	● Document and synthesize user-testing results. ● Prepare materials needed for user testing such as slides, consent forms, briefing materials.
Sensemaking, Evaluation, and Design Decisions *(Ipakita, Suriin, at Ayusin)*	evaluation in design as the process of interpreting feedback, identifying necessary improvements, and making justified design decisions.	32. organize and analyze feedback to identify patterns, priorities, actionable insights, tensions, gaps, and unresolved questions. 33. evaluate prototypes using established criteria such as context fit, feasibility, desirability, accessibility, ethics, safety, sustainability, and relevance. 34. explain trade-offs in design decisions by describing how different user needs are balanced, prioritized, or adjusted. 35. refine prototypes based on user	● Produce a feedback analysis report or matrix, documenting design decisions. ● Present updated prototypes.

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		feedback, reflections, and evaluation results.	
Capstone Communication, Reflection, Portfolio *(Komunikasyon, Pagninilay, at Portfolio sa Capstone)*	communication and reflection as integral to design work, demonstrating both process and learning beyond final outputs.	36. prepare a comprehensive capstone project report documenting the design and innovation process, including problem framing, ideation, prototyping, testing, and refinement. 37. present the capstone project using clear, organized, and audience-appropriate visual and oral communication strategies. 38. defend the capstone project by explaining its design, decisions, process, and outcomes before an audience or evaluation panel. 39. explain ethical, social, environmental, and sustainability considerations addressed in the project. 40. reflect on the design and innovation process to identify key learnings, strengths, limitations, and areas for improvement.	• Compile a design portfolio containing all documentation and outputs from the design process. • Deliver a final presentation or demonstration. • Write an individual reflection essay on the design process.

PERFORMANCE STANDARDS:

- The learners can demonstrate understanding of the design thinking process by developing a capstone project concept that offers solutions to real-world problems grounded in user stories, field observations, and contextual understanding. Working collaboratively in teams, they can establish shared norms and uphold ethical responsibilities when engaging with stakeholders.
- They can formulate a clear design challenge or problem statement, develop a documented design process portfolio, and produce at least one early testable prototype through analysis of user data, ideation, co-creation, and feedback from peers, teachers, and community members.
- They can also present a refined prototype or implemented solution, supported by evidence of user testing and iteration. They can

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	communicate their work through a comprehensive portfolio and organized presentation that showcase both the final output and the evolution of their understanding, including reflections on key learnings, limitations, and areas for further improvement.		